

The transverse z -integration of $d^3 N_2^{\text{NLO}}/d^3 k$

UV logarithms, their finite companions, and *all* the remaining terms

Group I, Row V (the $C_1 \otimes B_2$ two- ρ row) — complete separation at $z \rightarrow x$

0. What this note delivers

The transverse integrand of Eq. (1.1) is singular when $z \rightarrow x$. The object of the exercise is to isolate that singularity. The previous note stopped after extracting the divergent logarithm; everything else was dropped. Here *nothing* is dropped. The z -integral is performed in full and split into three groups:

$$\mathcal{W} \equiv \int d^2 r \, e^{iq \cdot r} \frac{T(r)}{D} = \underbrace{\mathcal{W}_{\text{UV}}}_{\text{the } \log(1/m^2)} + \underbrace{\mathcal{W}_{\text{fin}}}_{\text{finite companion of that log}} + \underbrace{\mathcal{W}_{\text{reg}}}_{\text{the eight structures that were never divergent}}$$

The three groups are given in closed form in §8. The logic is:

1. **An exact algebraic identity** (§4) $T(r) = C_{\text{UV}}(\xi) \frac{s \cdot P}{r^2} + T_{\text{fin}}(r)$, valid at every r , not just asymptotically. The first piece carries the whole $z \rightarrow x$ divergence; T_{fin} is integrable at $r = 0$. $C_{\text{UV}} = P_{gg}(\xi)/2N_c$.
2. **One divergent master integral** (§6), evaluated exactly — $\log$ *plus* its finite companion, with no expansion.
3. **Eight regular structures** (§7), each reduced to Bessel-function master integrals and written out.

Every step below is checked numerically (§12): the algebra symbolically against Eq. (1.1) itself, every master integral against direct two-dimensional quadrature, the assembled answer against a direct quadrature of the whole integrand, and the UV coefficient against the measured slope $d\mathcal{W}/d\log(1/m^2)$.

1. Variables, and the two typos in the source

With $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = p^+/k^+$, $\bar{\xi} = 1 - \xi$:

$$y - z = s - r, \quad x - z = -r, \quad (y - z)^2 = (r - s)^2, \quad (y - x)^2 = s^2.$$

The longitudinal denominator and the phase are

$$p^+(y - x)^2 + (k^+ - p^+)(y - z)^2 = k^+ \bar{\xi} D, \quad D = (r - s)^2 + M^2, \quad M^2 = \frac{\xi}{\bar{\xi}} s^2,$$

$$e^{-ik \cdot (w' - z)} e^{-i\xi k \cdot (z - x)} = e^{-ik \cdot w'} e^{ik \cdot x} e^{iq \cdot r}, \quad q \equiv \bar{\xi} k.$$

Two abbreviations are used throughout:

$$D_0 \equiv D|_{r=0} = s^2 + M^2 = \frac{s^2}{\bar{\xi}}, \quad q \equiv |q|.$$

The singularity of the problem sits at $r \rightarrow 0$, i.e. $z \rightarrow x$: that is where $1/r^2$ and $1/r^4$ blow up. There is a *second* place where denominators vanish, $r \rightarrow s$ i.e. $z \rightarrow y$; §3 shows it is harmless, and §10 shows what it does at $\xi \rightarrow 0$.

Two things in the source that had to be fixed before anything else.

(i) Eq. (1.8) is off by a factor 2 in one denominator. Contracting $-\frac{k^+-p^+}{p^+} \left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2} \right]$ with $\frac{P^i r^m}{P^2 r^2}$ gives $\frac{(P \cdot (s-r))}{P^2 r^2} \left[-1 + \frac{r \cdot s}{2s^2} \right] \cdot (-\bar{\xi}/\xi)$, i.e.

$$\Theta_4 = \frac{\bar{\xi}}{\xi} \frac{(s \cdot P) - (P \cdot r)}{P^2 r^2} \left[1 - \frac{r \cdot s}{2s^2} \right],$$

not $\left[1 - \frac{r \cdot s}{s^2} \right]$ as printed. This has been verified symbolically (§12A); Eqs. (1.4)–(1.7), (1.9), (1.10) are all correct as printed. The corrected Θ_4 is used below — it changes c_3 , c_4 , c_5 but *not* C_{UV} .

(ii) Two conventions in (1.1) vs. (1.11). (1.1) carries a phase $e^{-ik \cdot (z-x)}$ and a prefactor p^+/k^+ ; (1.3) and (1.11) require instead $e^{-i\xi k \cdot (z-x)}$ (otherwise no $e^{iq \cdot r}$ survives and the z -integral has nothing to converge on at large r) and no explicit p^+/k^+ . This note follows (1.3)/(1.11). The factor is overall and does not touch the z -integral; it only reweights the p^+ integral of §10, where the alternative is quoted.

2. The integrand in a nine-structure basis

Write the bracket of (1.1) as $T(r)/P^2$, so that T is the whole bracket with the universal $1/P^2$ stripped off. Contracting Eq. (1.1) structure by structure and collecting terms (all done symbolically, §12A) gives the compact statement

$$T(r) = \sum_{k=1}^9 c_k(\xi) S_k(r),$$

	structure $S_k(r)$	coefficient $c_k(\xi)$	behaviour as $r \rightarrow 0$	origin
S_1	$\frac{(P \cdot r)(s \cdot r)}{r^4}$	$2\xi\bar{\xi} [= d_\perp \xi\bar{\xi}]$	$\sim 1/r^2$ — UV	$\Theta_1, \Theta_3, \Theta_5, \Theta_6$
S_2	$\frac{s \cdot P}{r^2}$	$\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}$	$\sim 1/r^2$ — UV	Θ_2, Θ_4
S_3	$\frac{P \cdot r}{r^2}$	$\xi^2 - \frac{5}{2}\xi + 2 - \frac{1}{\xi}$	$\sim 1/r$ — integrable	all six
S_4	$\frac{(P \cdot r)(s \cdot r)}{s^2 r^2}$	$\frac{\bar{\xi}^2}{2\xi}$	$O(1)$	Θ_3, Θ_4
S_5	$\frac{(s \cdot P)(s \cdot r)}{s^2 r^2}$	$-\frac{1}{2\xi}$	$O(1)$	Θ_4, Θ_6
S_6	$\frac{s \cdot P}{s^2}$	$\frac{1}{2}$	$O(1)$	Θ_6
S_7	$\frac{(s \cdot P) r \cdot (r-s)}{r^2(r-s)^2}$	$-\frac{\xi}{2\xi}$	$\sim 1/r$ — integrable	Θ_2
S_8	$\frac{(P \cdot r) s \cdot (r-s)}{r^2(r-s)^2}$	$\frac{\xi}{2}$	$\sim 1/r$ — integrable	Θ_5
S_9	$\frac{(s \cdot r) P \cdot (r-s)}{r^2(r-s)^2}$	$-\frac{1}{2}$	$\sim 1/r$ — integrable	Θ_6

Two remarks on the table. First, c_1 is the only place the transverse dimension enters: $c_1 = d_\perp \xi\bar{\xi}$, from the instantaneous $\delta^{im} d_\perp$ term of Θ_1 , and $d_\perp = 2$ has been used elsewhere. (In $d_\perp = 2 - 2\epsilon$ one would keep $c_1 = d_\perp \xi\bar{\xi}$ and $c_3 \rightarrow c_3 + (2 - \frac{d_\perp}{2})\xi\bar{\xi}$.) Second, the coefficient of S_1 is *not* Θ_1 alone: Θ_1 gives $d_\perp \xi\bar{\xi}$, and $\Theta_3, \Theta_5, \Theta_6$ give $-2\bar{\xi} - 2\xi + 2 = 0$. Their cancellation is what the previous note meant by " T_3, T_5, T_6 drop out"; it is a statement about S_1 only, and those three structures certainly do *not* drop out of the rest of the answer.

3. Where the singularities are — power counting

3.1 $r \rightarrow 0$ ($z \rightarrow x$) — the UV

Near $r = 0$ one has $D \rightarrow D_0$ and $(r-s)^2 \rightarrow s^2$, both finite and nonzero, and $e^{iq \cdot r} \rightarrow 1$. In two dimensions $d^2 r = r dr d\theta$, so a term behaving like r^{-n} is log-divergent for $n = 2$, convergent for $n \leq 1$. Using $\langle \hat{r}^i \rangle = 0$ and $\langle \hat{r}^i \hat{r}^j \rangle = \delta^{ij}/2$:

$$\left\langle \frac{(P \cdot r)(s \cdot r)}{r^4} \right\rangle = \frac{s \cdot P}{2r^2}, \quad \left\langle \frac{s \cdot P}{r^2} \right\rangle = \frac{s \cdot P}{r^2}, \quad \left\langle \frac{P \cdot r}{r^2} \right\rangle = 0.$$

Only S_1 and S_2 survive. S_3, S_7, S_8, S_9 go like $1/r$ — odd, hence angular-averaging to zero at leading order, and in any case integrable; S_4, S_5, S_6 are bounded. So **the entire $z \rightarrow x$ divergence lives in S_1 and S_2** , with

$$\frac{c_1}{2} + c_2 = \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\xi} \equiv C_{UV}(\xi) = \frac{P_{gg}(\xi)}{2N_c}, \quad P_{gg} = 2N_c \left[\frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi) \right].$$

The one-to-one correspondence is: $\xi(1-\xi)$ from the instantaneous Θ_1 , $\xi/(1-\xi)$ from Θ_2 , $(1-\xi)/\xi$ from Θ_4 .

3.2 $r \rightarrow s$ ($z \rightarrow y$) — not a divergence

S_7, S_8, S_9 have $(r-s)^2$ in the denominator, but each numerator vanishes *linearly* there,

$$r \cdot (r-s) \Big|_{r \rightarrow s} = s \cdot (r-s) + O((r-s)^2), \quad s \cdot (r-s), \quad P \cdot (r-s),$$

so each behaves as $1/|r-s|$, which is integrable in two dimensions. $D = (r-s)^2 + M^2$ is regular there for $M^2 > 0$. Hence there is **no divergence at $z \rightarrow y$** and no second regulator is needed — unlike in the previous note, where an auxiliary μ^2 was introduced and its cancellation checked. Once $M^2 \rightarrow 0$, however ($\xi \rightarrow 0$), the $z \rightarrow y$ region does generate a $\log M^2$; this is worked out in §10.3 and, importantly, it *cancels* between the groups.

3.3 $r \rightarrow \infty$

S_6 carries no $1/r^2$ at all, so $S_6/D \sim r^{-2}$: the z -integral converges at large z only through the oscillation $e^{iq \cdot r}$. Several other terms are likewise only conditionally convergent. This is why the phase must be $e^{-i\xi k \cdot (z-x)}$ (see §1) — without $q = \bar{\xi}k$ the large- z end does not converge.

4. The exact UV/finite split of the integrand

The angular argument of §3.1 identifies the divergence but is asymptotic. It can be turned into an *exact identity*. Split S_1 into its trace and its traceless part,

$$\frac{(P \cdot r)(s \cdot r)}{r^4} = \underbrace{\frac{s \cdot P}{2r^2}}_{\text{trace}} + \underbrace{P_i s_j \left[\frac{r^i r^j}{r^4} - \frac{\delta^{ij}}{2r^2} \right]}_{\text{traceless}}.$$

The traceless bracket is orthogonal to every function of r^2 alone, so it contributes no logarithm: its z -integral converges at $r = 0$. Since $\frac{c_1}{2} + c_2 = C_{UV}$, the trace piece of S_1 and all of S_2 combine into a single term and

$$\begin{aligned} T(r) &= C_{UV}(\xi) \frac{s \cdot P}{r^2} + T_{\text{fin}}(r), \quad C_{UV}(\xi) = \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}(\xi)}{2N_c}, \\ T_{\text{fin}}(r) &= 2\xi \bar{\xi} \left[\frac{(P \cdot r)(s \cdot r)}{r^4} - \frac{s \cdot P}{2r^2} \right] \quad (\text{traceless, UV finite}) \\ &\quad + \left(\xi^2 - \frac{5}{2}\xi + 2 - \frac{1}{\xi} \right) \frac{P \cdot r}{r^2} + \frac{\bar{\xi}^2}{2\xi} \frac{(P \cdot r)(s \cdot r)}{s^2 r^2} - \frac{1}{2\xi} \frac{(s \cdot P)(s \cdot r)}{s^2 r^2} + \frac{1}{2} \frac{s \cdot P}{s^2} \\ &\quad - \frac{\xi}{2\bar{\xi}} \frac{(s \cdot P) r \cdot (r-s)}{r^2 (r-s)^2} + \frac{\xi}{2} \frac{(P \cdot r) s \cdot (r-s)}{r^2 (r-s)^2} - \frac{1}{2} \frac{(s \cdot r) P \cdot (r-s)}{r^2 (r-s)^2}. \end{aligned}$$

This is the central algebraic statement of the note, and it is an identity in r — verified symbolically against Eq. (1.1) itself (§12A). Everything after this point is just the evaluation of two-dimensional Fourier integrals. Note that T_{fin} is *not* small: it contains $1/\xi$ and $1/\bar{\xi}$ poles of its own, and at $\xi \rightarrow 1$ it does not vanish (§10.4).

5. Master integrals

Every term of T/D is of the form $N(r)/[(r^2)^a (r-s)^{2b} D]$ with $a \in \{0, 1, 2\}$, $b \in \{0, 1\}$ and $N \in \{1, r^i, r^i r^j\}$. Since $(r-s)^2$ and $D = (r-s)^2 + M^2$ differ only by a constant, partial fractions

$$\frac{1}{(r-s)^2 D} = \frac{1}{M^2} \left[\frac{1}{(r-s)^2} - \frac{1}{D} \right]$$

reduce everything to two-denominator integrals: the pair $\{r^2, D\}$ and the massless pair $\{r^2, (r-s)^2\}$.

5.1 Feynman parametrisation

With $A = r^2 + m^2$ (m the UV regulator, needed only for the divergent master) and $B = (r-s)^2 + \Delta^2$,

$$\alpha A + (1-\alpha)B = (r-b)^2 + \Delta, \quad b = \bar{\alpha}s, \quad \Delta(\alpha) = \bar{\alpha}(\alpha s^2 + \Delta^2) + \alpha m^2, \quad \bar{\alpha} \equiv 1 - \alpha,$$

$$\frac{1}{AB} = \int_0^1 \frac{d\alpha}{[\alpha A + \bar{\alpha} B]^2}, \quad \frac{1}{A^2 B} = \int_0^1 \frac{2\alpha d\alpha}{[\alpha A + \bar{\alpha} B]^3}.$$

For the pair $\{r^2, D\}$ take $\Delta^2 = M^2$; for the massless pair $\{r^2, (r-s)^2\}$ take $\Delta^2 = m^2 = 0$, i.e. $\Delta_0(\alpha) = \alpha \bar{\alpha} s^2$.

5.2 The two-dimensional Bessel transforms

Shift $r = b + u$; the phase factorises, $e^{iq \cdot r} = E e^{iq \cdot u}$ with $E \equiv e^{i\bar{\alpha} q \cdot s}$. Everything then follows from the 2D Yukawa transform and two derivative rules, with $\lambda \equiv q\sqrt{\Delta}$:

$$U_1 = \int \frac{d^2 u e^{iq \cdot u}}{u^2 + \Delta} = 2\pi K_0(\lambda), \quad U_{n+1} = -\frac{1}{n} \frac{\partial U_n}{\partial \Delta}, \quad V_n^i = -i \frac{\partial U_n}{\partial q_i}, \quad T_n^{ij} = -\frac{\partial^2 U_n}{\partial q_i \partial q_j}.$$

Using $K'_0 = -K_1$ and $K'_1 = -K_0 - K_1/x$, and $\frac{d}{d\Delta} [\sqrt{\Delta} K_1(q\sqrt{\Delta})] = -\frac{q}{2} K_0$:

$$U_2 = \frac{\pi q}{\sqrt{\Delta}} K_1(\lambda), \quad U_3 = \frac{\pi q}{2} \left[\frac{K_1(\lambda)}{\Delta^{3/2}} + \frac{q K_0(\lambda)}{2\Delta} \right],$$

$$V_2^i = i\pi q^i K_0(\lambda), \quad V_3^i = \frac{i\pi q^i q}{4\sqrt{\Delta}} K_1(\lambda),$$

$$T_2^{ij} = \pi \left[\delta^{ij} K_0(\lambda) - \frac{q^i q^j \sqrt{\Delta}}{q} K_1(\lambda) \right], \quad T_3^{ij} = \frac{\pi}{4} \left[\frac{\delta^{ij} q}{\sqrt{\Delta}} K_1(\lambda) - q^i q^j K_0(\lambda) \right].$$

5.3 The masters

Collecting, with $b = \bar{\alpha}s$ and $E = e^{i\bar{\alpha} q \cdot s}$, the six masters that occur are (all UV finite except $\mathcal{G}$):

$$\begin{aligned} \Phi &= \int d^2 r \frac{e^{iq \cdot r}}{D} = 2\pi e^{iq \cdot s} K_0(Mq) \quad (\text{closed form, no parameter integral}), \\ \Psi^i &= \int d^2 r \frac{e^{iq \cdot r} r^i}{r^2 D} = \pi \int_0^1 d\alpha E \left[\bar{\alpha} s^i \frac{q}{\sqrt{\Delta}} K_1(\lambda) + i q^i K_0(\lambda) \right], \\ \mathcal{N}^{ij} &= \int d^2 r \frac{e^{iq \cdot r} r^i r^j}{r^2 D} = \int_0^1 d\alpha E \left[b^i b^j U_2 + b^i V_2^j + b^j V_2^i + T_2^{ij} \right], \\ \mathcal{T}^{ij} &= \int d^2 r e^{iq \cdot r} \frac{r^i r^j - \frac{1}{2} \delta^{ij} r^2}{r^4 D} = \int_0^1 d\alpha E \left\{ 2\alpha [b^i b^j U_3 + b^i V_3^j + b^j V_3^i + T_3^{ij}] \right\}_{\text{traceless}}, \\ \mathcal{G} &= \int d^2 r \frac{e^{iq \cdot r}}{(r^2 + m^2) D} = \int_0^1 d\alpha E U_2(\Delta), \quad \Delta = \bar{\alpha}(\alpha s^2 + M^2) + \alpha m^2. \end{aligned}$$

In $\Phi, \Psi, \mathcal{N}, \mathcal{T}$ one may set $m = 0$, so $\Delta = \bar{\alpha}(\alpha s^2 + M^2)$. Three checks that the reader can make immediately: $\mathcal{N}^{ii} = \Phi$ (because $r^i r^i / r^2 = 1$); $\mathcal{T}^{ii} = 0$ by construction; and $\mathcal{T}$ is finite at $\alpha \rightarrow 1$ only *after* the traceless projection — the divergent piece of $2\alpha T_3^{ij}$ is pure δ^{ij} . Taking the trace *before* projecting is the one trap here: $\int \frac{r^i r^j}{(r^2 + m^2)^2 D}$ has trace $\mathcal{G} - m^2 \int \frac{1}{(r^2 + m^2)^2 D}$, and the second term does *not* vanish as $m \rightarrow 0$ — it tends to π/D_0 .

5.4 The massless masters

For the pair $\{r^2, (r-s)^2\}$, with $\Delta_0 = \alpha \bar{\alpha} s^2$, $\lambda_0 = qs\sqrt{\alpha \bar{\alpha}}$, $r = u + \bar{\alpha}s$ and $r-s = u - \alpha s$:

$$\begin{aligned}
A_7 &= \int d^2r e^{iq \cdot r} \frac{r \cdot (r-s)}{r^2(r-s)^2} = \pi \int_0^1 d\alpha E \left[2K_0(\lambda_0) - 2\lambda_0 K_1(\lambda_0) + i(1-2\alpha)(q \cdot s) K_0(\lambda_0) \right], \\
A_8 &= \int d^2r e^{iq \cdot r} \frac{(P \cdot r) s \cdot (r-s)}{r^2(r-s)^2} = \int_0^1 d\alpha E \left[P_i s_j T_2^{ij} - \alpha s^2 (P \cdot V_2) + \bar{\alpha} (P \cdot s) (s \cdot V_2) - \alpha \bar{\alpha} s^2 (P \cdot s) U_2 \right], \\
A_9 &= \int d^2r e^{iq \cdot r} \frac{(s \cdot r) P \cdot (r-s)}{r^2(r-s)^2} = \int_0^1 d\alpha E \left[s_i P_j T_2^{ij} - \alpha (P \cdot s) (s \cdot V_2) + \bar{\alpha} s^2 (P \cdot V_2) - \alpha \bar{\alpha} s^2 (P \cdot s) U_2 \right].
\end{aligned}$$

(A_7 used $u^2 = (u^2 + \Delta_0) - \Delta_0$, so that $\int u^2 e^{iqu}/(u^2 + \Delta_0)^2 = U_1 - \Delta_0 U_2$, and $\Delta_0 = \alpha \bar{\alpha} s^2$ collapses the two U_2 terms into $-2\lambda_0 K_1$.) These are ξ -independent: they depend on q and s only. A useful identity: $A_8 = A_9$ when $P = s$, since then the two numerators coincide.

6. The divergent master, exactly — log and companion

There is no need to expand anything. Use the identity $\frac{1}{r^2 D} = \frac{1}{r^2 D_0} + \frac{1}{r^2} \left(\frac{1}{D} - \frac{1}{D_0} \right)$ together with

$$\frac{1}{D} - \frac{1}{D_0} = \frac{D_0 - D}{D D_0} = \frac{2r \cdot s - r^2}{D D_0} \implies \frac{1}{r^2} \left(\frac{1}{D} - \frac{1}{D_0} \right) = \frac{1}{D_0} \left[\frac{2r \cdot s}{r^2 D} - \frac{1}{D} \right],$$

which is regular at $r = 0$ (the bracket vanishes linearly there), so m may be set to zero in it. Hence, exactly,

$$\begin{aligned}
\mathcal{G} &= \int \frac{d^2r e^{iq \cdot r}}{(r^2 + m^2) D} = \frac{2\pi}{D_0} K_0(mq) + \frac{1}{D_0} \left[2s \cdot \Psi - \Phi \right] + O(m^2), \\
2\pi K_0(mq) &= \pi \log \frac{4e^{-2\gamma_E}}{m^2 q^2} = \underbrace{\pi \log \frac{D_0}{m^2}}_{\text{the UV log}} + \underbrace{\pi \log \frac{4e^{-2\gamma_E}}{q^2 D_0}}_{\text{finite}}.
\end{aligned}$$

The second line is a *choice* of where to cut the logarithm; the argument $D_0 = s^2/\bar{\xi}$ is picked so that the UV log reads $\log[(y-x)^2/\bar{\xi} m^2]$, which is the combination that pairs with the running coupling. Any other choice moves a constant between $\mathcal{W}_{\text{UV}}$ and $\mathcal{W}_{\text{fin}}$ and changes neither separately-meaningful physics nor the sum (see §11).

Dimensional regularisation. The whole m -dependence is the single factor $\pi \log(1/m^2)$. In $d_\perp = 2 + 2\epsilon_{\text{UV}}$, $\int d^{d_\perp} r e^{iq \cdot r} / r^2 = \pi^{1+\epsilon} 2^{2\epsilon} \Gamma(\epsilon) (q^2)^{-\epsilon}$, and matching the two regulators gives the dictionary

$$\log \frac{1}{m^2} \longleftrightarrow \frac{1}{\epsilon_{\text{UV}}} + \log(\pi e^{\gamma_E} \mu^2),$$

i.e. $\pi \log(1/m^2) \rightarrow \pi/\epsilon_{\text{UV}} + \dots$, the pole that cancels against the $-2/\epsilon$ of the gluon self-energy counterterm. (With the opposite convention $d_\perp = 2 - 2\epsilon$ the pole is $-1/\epsilon$.)

7. The eight regular structures, reduced

Each surviving structure maps onto the masters of §5. The three $(r-s)^2$ structures are handled by the partial fraction of §5, which turns $\int N/[r^2(r-s)^2 D]$ into $\frac{1}{M^2} \left[\int \frac{N}{r^2(r-s)^2} - \int \frac{N}{r^2 D} \right]$:

$$\begin{aligned}
\mathcal{S}_7 &\equiv \int d^2r e^{iq \cdot r} \frac{(s \cdot P) r \cdot (r-s)}{r^2(r-s)^2 D} = \frac{s \cdot P}{M^2} \left[A_7 - \Phi + s \cdot \Psi \right], \\
\mathcal{S}_8 &\equiv \int d^2r e^{iq \cdot r} \frac{(P \cdot r) s \cdot (r-s)}{r^2(r-s)^2 D} = \frac{1}{M^2} \left[A_8 - P_i s_j \mathcal{N}^{ij} + s^2 (P \cdot \Psi) \right], \\
\mathcal{S}_9 &\equiv \int d^2r e^{iq \cdot r} \frac{(s \cdot r) P \cdot (r-s)}{r^2(r-s)^2 D} = \frac{1}{M^2} \left[A_9 - s_i P_j \mathcal{N}^{ij} + (P \cdot s) (s \cdot \Psi) \right],
\end{aligned}$$

using $r \cdot (r-s) = r^2 - r \cdot s$, $s \cdot (r-s) = s \cdot r - s^2$, $P \cdot (r-s) = P \cdot r - P \cdot s$ and $\int e^{iq \cdot r} r^2 / (r^2 D) = \Phi$. The apparent $1/M^2$ poles are spurious: at $M^2 = 0$ one has $D = (r-s)^2$ and the bracket vanishes identically, so each $\mathcal{S}_k$ is finite as $\xi \rightarrow 0$.

structure	coefficient	z -integral $\int d^2r e^{iq \cdot r} S_k / D$
$S_1 - \frac{1}{2} S_2$ (traceless)	$2\xi\bar{\xi}$	$P_i s_j \mathcal{T}^{ij}$
S_2 (trace + S_2)	C_{UV}	$(s \cdot P) \mathcal{G}$ — the only divergent one
S_3	$\xi^2 - \frac{5}{2}\xi + 2 - \frac{1}{\xi}$	$P \cdot \Psi$
S_4	$\bar{\xi}^2 / 2\xi$	$P_i s_j \mathcal{N}^{ij} / s^2$
S_5	$-1/2\xi$	$(s \cdot P)(s \cdot \Psi) / s^2$
S_6	$1/2$	$(s \cdot P) \Phi / s^2$
S_7, S_8, S_9	$-\frac{\xi}{2\bar{\xi}}, \frac{\xi}{2}, -\frac{1}{2}$	$\mathcal{S}_7, \mathcal{S}_8, \mathcal{S}_9$ as above

8. The result

With $q = \bar{\xi}k$, $D_0 = s^2/\bar{\xi}$, $M^2 = (\xi/\bar{\xi})s^2$, $C_{UV} = P_{gg}/2N_c$:

$$\mathcal{W} = \int d^2r e^{iq \cdot r} \frac{T(r)}{D} = \mathcal{W}_{UV} + \mathcal{W}_{fin} + \mathcal{W}_{reg},$$

$$\text{(I) UV log:} \quad \mathcal{W}_{UV} = C_{UV}(\xi) (s \cdot P) \frac{\pi\bar{\xi}}{s^2} \log \frac{s^2}{\bar{\xi} m^2},$$

$$\text{(II) its finite companion:} \quad \mathcal{W}_{fin} = C_{UV}(\xi) (s \cdot P) \frac{\bar{\xi}}{s^2} \left[\pi \log \frac{4e^{-2\gamma_E} \bar{\xi}}{q^2 s^2} + 2 s \cdot \Psi - \Phi \right],$$

$$\begin{aligned} \text{(III) the other terms:} \quad \mathcal{W}_{reg} = & 2\xi\bar{\xi} P_i s_j \mathcal{T}^{ij} + \left(\xi^2 - \frac{5}{2}\xi + 2 - \frac{1}{\xi} \right) P \cdot \Psi + \frac{\bar{\xi}^2}{2\xi s^2} P_i s_j \mathcal{N}^{ij} \\ & - \frac{(s \cdot P)(s \cdot \Psi)}{2\xi s^2} + \frac{(s \cdot P)\Phi}{2s^2} - \frac{\xi}{2\bar{\xi}} \mathcal{S}_7 + \frac{\xi}{2} \mathcal{S}_8 - \frac{1}{2} \mathcal{S}_9. \end{aligned}$$

Group (I) is the only place m (or $1/\epsilon_{UV}$) appears. Groups (II) and (III) are finite, and are given entirely by the six masters of §5 — one closed form (Φ) and five single α -integrals of Bessel functions. Nothing has been expanded, approximated or discarded anywhere between §2 and here.

9. Back to the cross section

Inserting into (1.11) — the bracket is T/P^2 , so the z -integral of the bracket is $\mathcal{W}/P^2$ —

$$\frac{d^3 N_2^{\text{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4 k^+} \int_{x,y,x',w'} e^{-ik \cdot w'} e^{ik \cdot x} \int_{\Lambda/k^+}^{1-\Lambda/k^+} \frac{d\xi}{2\pi} \frac{1}{\bar{\xi}} \frac{\mathcal{W}_{UV} + \mathcal{W}_{fin} + \mathcal{W}_{reg}}{P^2} [\dots] \rho^{b'}(x') \rho^f(y),$$

where $[\dots]$ is the colour structure of (1.1). For group (I) the explicit $\bar{\xi}$ of $\mathcal{W}_{UV}$ cancels the $1/\bar{\xi}$ exactly and one is left with a product of two Weizsäcker–Williams kernels times a logarithm of the dipole size:

$$\frac{1}{\bar{\xi}} \frac{\mathcal{W}_{UV}}{P^2} = \pi \frac{P_{gg}(\xi)}{2N_c} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \log \frac{(y - x)^2}{\bar{\xi} m^2},$$

which is the object that pairs with the $-2/\epsilon$ of $A_{N_c}^{(3)}$ and becomes $\alpha_s((y - x)^{-2})$. Groups (II) and (III) carry a different transverse structure — they are *not* proportional to the WW×WW kernel — and this is precisely the content that was missing before.

10. The p^+ integration

10.1 Group (I): P_{gg} , and b_0

Write $dp^+ = k^+ d\xi$, $\lambda = \Lambda/k^+$, $\ell_k = \log(k^+/\Lambda)$, $L = \log[(y - x)^2/m^2]$; the logarithm of §9 is $L + \log \frac{1}{1-\xi}$. All six integrals are elementary:

piece of $P_{gg}/2N_c$	from	$\int_{\lambda}^{1-\lambda} d\xi$	$\int_{\lambda}^{1-\lambda} d\xi \log \frac{1}{1-\xi}$
$\xi(1-\xi)$	Θ_1 (instantaneous)	$\frac{1}{6}$	$\frac{5}{36}$
$\frac{\xi}{1-\xi}$	Θ_2	$\ell_k - 1$	$\frac{1}{2}\ell_k^2 - 1$
$\frac{1-\xi}{\xi}$	Θ_4	$\ell_k - 1$	$\frac{\pi^2}{6} - 1$
total		$2\ell_k - \frac{11}{6}$	$\frac{1}{2}\ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36}$

$$\int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{2\pi} \frac{P_{gg}(\xi)}{2N_c} \log \frac{k^+(y-x)^2}{(k^+-p^+)m^2} = \frac{k^+}{2\pi} \left[\left(2\ell_k - \frac{11}{6}\right) \log \frac{(y-x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} \right].$$

Three readings. The double logarithm comes only from Θ_2 and the $\pi^2/6$ only from Θ_4 ; it is the instantaneous Θ_1 that turns -2 into $-\frac{11}{6}$ and -2 into $-\frac{67}{36}$. And $-\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b_0$ at $N_f = 0$: this row supplies, on its own, the $\frac{11N_c}{6}\delta(1-\xi)$ of the regularised splitting function, hence part of the running coupling.

If the p^+/k^+ of (1.1) is kept (see §1(ii)), every weight acquires an extra ξ and the plain integrals become $\int \xi \cdot \frac{1-\xi}{\xi} = \frac{1}{2}$, $\int \xi \cdot \frac{\xi}{1-\xi} = \ell_k - \frac{3}{2}$, $\int \xi \cdot \xi(1-\xi) = \frac{1}{12}$, total $\ell_k - \frac{11}{12}$ — one rapidity logarithm instead of two, and no b_0 . Which convention is intended should be settled before this row is used numerically.

10.2 Groups (II) and (III): why their ξ -integral is not elementary

$q = \bar{\xi}k$ depends on ξ , and so do $M^2 = (\xi/\bar{\xi})s^2$ and $D_0 = s^2/\bar{\xi}$. The masters therefore depend on ξ both through their coefficients and through their arguments, and $\int d\xi$ cannot be done in closed form. What *can* be established is the endpoint behaviour, which is what decides the rapidity logarithms.

10.3 $\xi \rightarrow 0$: a $z \rightarrow y$ logarithm that cancels between (II) and (III)

As $\xi \rightarrow 0$, $M^2 \rightarrow 0$ and D develops a genuine singularity at $r = s$. For any smooth f , $\int d^2r f(r)/[(r-s)^2 + M^2] = \pi f(s) \log(1/M^2) + \dots$, so

$$\Psi^i \Big|_{\log} = \pi e^{iq \cdot s} \frac{s^i}{s^2} \Lambda_M, \quad \mathcal{N}^{ij} \Big|_{\log} = \pi e^{iq \cdot s} \frac{s^i s^j}{s^2} \Lambda_M, \quad \Phi \Big|_{\log} = \pi e^{iq \cdot s} \Lambda_M, \quad \mathcal{T}^{ij} \Big|_{\log} = \pi e^{iq \cdot s} \left[\frac{s^i s^j}{s^4} - \frac{\delta^{ij}}{2s^2} \right] \Lambda_M,$$

with $\Lambda_M \equiv \log(1/M^2) = \log(\bar{\xi}/\xi s^2)$. For $\mathcal{S}_{7,8,9}$ the numerators vanish at $r = s$, and the logarithm comes only from the $e^{iq \cdot p}$ expansion, $\mathcal{S}_k|_{\log} = i\pi e^{iq \cdot s} (q \cdot c_k) \Lambda_M / 2s^2$ with $c_7 = (s \cdot P)s$, $c_8 = (P \cdot s)s$, $c_9 = s^2 P$ — all $O(1)$, not $O(1/\xi)$. Keeping only the $1/\xi$ poles ($c_3 \rightarrow -\frac{1}{\xi}$, $c_4 \rightarrow \frac{1}{2\xi}$, $c_5 \rightarrow -\frac{1}{2\xi}$; sum $-\frac{1}{\xi}$):

$$\mathcal{W}_{\text{reg}} \xrightarrow{\xi \rightarrow 0} -\frac{\pi}{\xi} e^{iq \cdot s} \frac{s \cdot P}{s^2} \Lambda_M + O\left(\frac{1}{\xi}\right), \quad \mathcal{W}_{\text{fin}} \xrightarrow{\xi \rightarrow 0} +\frac{\pi}{\xi} e^{iq \cdot s} \frac{s \cdot P}{s^2} \Lambda_M + O\left(\frac{1}{\xi}\right),$$

because $C_{\text{UV}} \rightarrow 1/\xi$, $D_0 \rightarrow s^2$ and $(2s \cdot \Psi - \Phi)|_{\log} = (2-1)\pi e^{iq \cdot s} \Lambda_M$. The two cancel exactly. Both statements are confirmed numerically to five digits (§12E), where $\xi(\mathcal{W}_{\text{fin}} + \mathcal{W}_{\text{reg}})$ is seen to approach a constant while each piece separately grows like $\log(1/\xi)$.

Consequence. The $z \rightarrow y$ (collinear) logarithm is an artefact of where the UV cut is placed; it is absent from the sum. What survives at $\xi \rightarrow 0$ is a simple pole $\propto 1/\xi$, i.e. *one* rapidity logarithm ℓ_k from the finite and regular terms together — in addition to the ones carried by P_{gg} in group (I).

10.4 $\xi \rightarrow 1$: the regular block does not vanish

At fixed q , as $\bar{\xi} \rightarrow 0$ one has $M^2 \rightarrow \infty$: every master falls off like $1/M^2$ and the only survivor is $-\frac{\xi}{2\bar{\xi}} \mathcal{S}_7 \rightarrow -\frac{\xi}{2\bar{\xi}} \cdot \frac{(s \cdot P)A_7}{M^2}$, so

$$\lim_{\xi \rightarrow 1} \mathcal{W}_{\text{reg}} \Big|_{q \text{ fixed}} = -\frac{(s \cdot P)}{2s^2} A_7(q, s) \quad (\text{verified to 6 digits, §12F}).$$

With the physical $q = \bar{\xi}k$ the two limits do not commute — $A_7 \sim 2\pi \log(1/qs)$ as $q \rightarrow 0$ and the other masters grow too — but the conclusion is the same and is what matters: $\mathcal{W}_{\text{reg}}$ tends to a **finite nonzero constant** at the endpoint (numerically $+0.868\dots$ for the test kinematics), so the explicit $1/\bar{\xi}$ of the cross section turns it into a rapidity logarithm ℓ_k with a

coefficient that is *not* contained in P_{gg} . Any rapidity subtraction applied to this row must therefore act on the regular terms as well, not only on group (I).

11. What the split does and does not mean

- **Unambiguous.** The coefficient of $\log(1/m^2)$ — equivalently of $1/\epsilon_{UV}$ — is $C_{UV}(\xi)(s \cdot P)\pi\bar{\xi}/s^2$, i.e. $P_{gg}/2N_c$ times two WW kernels. So is the algebraic identity of §4, and so is the sum $\mathcal{W}_{UV} + \mathcal{W}_{fin} + \mathcal{W}_{reg}$.
- **Scheme-dependent.** The boundary between (I) and (II) is the argument chosen for the logarithm, here $D_0 = s^2/\bar{\xi}$. Choosing q^{-2} or s^2 instead moves constants (and, at $\xi \rightarrow 1$, a $\log \bar{\xi}$) between the two. The boundary between (II) and (III) is fixed by the traceless projection of §4, which is natural but not forced.
- **The $U(z)$ caveat.** In (1.1) the colour structure contains $U^{ce}(z)$, so the z -integral is not literally separable from it; (1.11) writes it as if it were. For group (I) this is harmless and is in fact the point: the divergence is local, $z \rightarrow x$, so $U^{ce}(z) \rightarrow U^{ce}(x)$ there and the coefficient multiplies a genuinely local colour structure — which is why it can be absorbed into the running coupling. For groups (II) and (III) the results above are the z -integral *at fixed colour structure*, i.e. they are exact if $U^{ce}(z)$ is taken outside, and otherwise they are the decomposition of the z -integrand, to be convolved with $U^{ce}(z)$. The algebraic split of §4 is unaffected either way, since it is pointwise in r .
- $d_\perp$. Only c_1 (hence the $\xi(1 - \xi)$ of P_{gg}) knows about the transverse dimension. If the transverse plane is continued to $2 - 2\epsilon$ to regulate, $c_1 = d_\perp \xi \bar{\xi}$ and the $O(\epsilon)$ part of c_1 multiplies the $1/\epsilon$ pole, producing an extra finite $-\xi \bar{\xi} \times (\text{constant})$ that belongs in group (II).

12. Numerical verification

Everything above is checked. The algebra is checked symbolically (sympy) directly against Eq. (1.1); the integrals are checked against direct two-dimensional quadrature of $\int d^2r e^{iq \cdot r} F(r)$, done in polar coordinates with q rotated onto the x -axis: periodic-trapezoid in the angle, adaptive quadrature in the radius out to $R_n = R_0 + n\pi/q$, and repeated averaging of the partial sums to kill the slowly decaying oscillatory tail.

A. the algebra (sympy, exact)

```

Theta_1 == (1.4)           : True
Theta_2 == (1.5)           : True
Theta_3 == (1.6)/(1.7)     : True
Theta_4 == (1.8) as printed : False    <-- the factor-2 typo
Theta_4 == (1.8) with s^2 -> 2 s^2 : True
Theta_5 == (1.9)           : True
Theta_6 == (1.10)          : True
T (d_perp=2) == sum_k c_k S_k : True
T - C_UV (s.P)/r^2 == T_fin : True

```


B–G. the integrals

```

kinematics:  s = [ 0.7 -0.4]  q = [0.9 0.6]  P = [1.3 0.5]  xi = 0.3
              s^2 = 0.6500  M^2 = 0.278571  D0 = 0.928571  s.P = 0.7100  q.s = 0.3900

A. master integrals:  alpha-representation  vs  direct 2D quadrature
      alpha-representation          2D quadrature          rel.diff
Psi^0      +2.7273301853+3.3432229840j      +2.7273301436+3.3432230085j      1.1e-08
Psi^1      -2.1029242522+1.3184267910j      -2.1029242881+1.3184267937j      1.4e-08
N^00       +2.0493314046+2.0734030364j      +2.0493311443+2.0734031386j      9.6e-08
N^01       -1.9056399937-0.3615375386j      -1.9056401726-0.3615376223j      1.0e-07
N^11       +2.6969517015-0.1224200369j      +2.6969516329-0.1224201107j      3.7e-08
Tt^00      +0.2544265956+0.9149478158j      +0.2544265787+0.9149478226j      1.9e-08
Tt^01      -1.4224659992-0.1649205388j      -1.4224660382-0.1649205438j      2.7e-08
A7         +2.4533167153+0.4845541087j      +2.4533161332+0.4845547275j      3.4e-07
A8         -0.3592796084-1.4936545930j      -0.3592788373-1.4936536083j      8.1e-07
A9         -0.9001711373+1.2449007205j      -0.9007825997+1.2446497402j      4.3e-04

B. exact identities
   trace N^ij - Phi                      : 4.58e-15
   trace Tt^ij                          : 1.39e-15
   G(m) alpha-rep vs (2pi/D0)K0(mq)+(2s.Psi-Phi)/D0, m=1e-3 : 5.60e-07
   A8(P=s) - A9(P=s)                    : 0.00e+00

C. the regular block W_reg (assembled from masters) vs direct quadrature of T_fin/D
   xi=0.30  assembled -4.48077798-10.60311732j      quad -4.47731510-10.60169400j      rel
3.3e-04
   xi=0.65  assembled -0.48091094-2.99952516j      quad -0.48083411-2.99949381j      rel
2.7e-05

D. the UV coefficient: d W / d log(1/m^2) of the *regulated original* integrand
   xi=0.30  measured +7.137881  predicted C_UV (s.P) pi/D0 = +7.138842  rel 1.3e-04
   xi=0.65  measured +3.150126  predicted C_UV (s.P) pi/D0 = +3.150493  rel 1.2e-04

E. xi -> 0 : the ln(1/M^2) of the companion cancels that of the regular block
   xi      xi*W_companion      xi*W_reg      xi*(W_comp+W_reg)
1e-02     +10.6714+5.5627j      -13.1341-9.2589j      -2.462678-3.696233j
1e-03     +18.1362+8.6060j      -20.6686-12.3818j      -2.532414-3.775854j
1e-04     +25.4732+11.6164j      -28.0125-15.4003j      -2.539311-3.783930j
1e-05     +32.7856+14.6213j      -35.3256-18.4061j      -2.539999-3.784738j
1e-06     +40.0943+17.6255j      -42.6344-21.4103j      -2.540068-3.784819j
coefficient of ln(1/M^2)/xi in W_reg: predicted -3.173905-1.304649j
                                         measured -3.174994-1.305007j  ratio 1.00033

F. xi -> 1 at fixed q : W_reg -> -(s.P) A7 /(2 s^2)
   prediction -1.3398884-0.2646411j
   xibar=1e-03  W_reg = -1.3394803-0.2763443j
   xibar=1e-04  W_reg = -1.3398468-0.2658110j
   xibar=1e-05  W_reg = -1.3398842-0.2647581j

G. the p+ integrals of P_gg/(2Nc) (mpmath, 30 digits, lambda=1e-8)
   sum of plain integrals 35.00802817457  2 l_k - 11/6 = 35.00802815457
   sum of log-weighted ones 169.4445626753  l_k^2/2+pi^2/6-67/36 = 169.444562491

```

Comments on the table. A_9 is the slowest-converging quadrature (its integrand falls only as r^{-2} with a nonzero angular average); pushing the averaging depth from 14 to 20 levels moves the quadrature from -0.90598 to -0.90078 against the α -representation's -0.90017 , i.e. it is converging onto it. Row D measures $dW/d\log(1/m^2)$ on the *original* integrand with $r^2 \rightarrow r^2 + m^2$ everywhere — a regularisation different from the one used in §6, which is the point: the log coefficient does not care.

13. Summary card

Integrand $T(r) = C_{UV}(\xi) \frac{s \cdot P}{r^2} + T_{fin}(r)$, $C_{UV} = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}}{2N_c}$ (exact, pointwise).

z-integral $\mathcal{W} = \mathcal{W}_{UV} + \mathcal{W}_{fin} + \mathcal{W}_{reg}$ of §8, built from Φ , Ψ^i , $\mathcal{N}^{ij}$, $\mathcal{T}^{ij}$, $A_{7,8,9}$ and $\mathcal{S}_{7,8,9}$.

UV $\frac{1}{\bar{\xi}} \frac{\mathcal{W}_{UV}}{P^2} = \pi \frac{P_{gg}(\xi)}{2N_c} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \log \frac{(y - x)^2}{\bar{\xi} m^2}$, $\int \frac{dp^+}{2\pi} \rightarrow \frac{k^+}{2\pi} \left[(2\ell_k - \frac{11}{6})L + \frac{1}{2}\ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36} \right]$.

Everything else finite; carries a different transverse structure; has its own $1/\xi$ and $1/\bar{\xi}$ endpoint behaviour, hence its own rapidity logarithms (§10.3–10.4); and its $\log M^2$ cancels between groups (II) and (III).

Code

zint/src/algebra.py — sympy reduction of Eq. (1.1) to $T(r)$, the nine-structure basis and the UV split (§12A).
zint/src/masters.py — the α -representations of §5. zint/src/quad2d.py — the oscillatory 2D quadrature.
zint/src/assemble.py — the assembly of §8. zint/src/run_all.py — every number in §12, written to zint/RESULTS.txt.